

ELECENG 2EI4

Project 5: Digital-to-Analog Converter

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1. Summary and Design

1.1 Summary

Digital-to-Analog Converters (DAC) are circuits that are capable of assigning an analog voltage or current to a digital code. In embedded technologies and computer systems, the digital-to-analog converter acts as an interface between the digital computing device and analog output signals. DACs are very commonly used in music and audio technology where converting digital audio files to analog voltage signals are essential to be played on speakers and other audio emitting devices. DACs are also used in sensor and actuator networks to precisely generate voltage levels required for accurate control and measurements.

This project involves the design, simulation, and implementation of a 3-bit DAC that is capable of producing an output voltage can convert signal that is within the range of 0-5V, with a true step size of 0.714V. This was done using an operational amplifier configured as a summing amplifier.

Experimental measurements were then conducted to analyse and characteristics of the DAC including the gain error, offset error and differential non-linearity (DNL) of the circuit. The results demonstrate that the circuit produces a near-linear response, with small variations from ideal behaviour, primarily due to the op-amps non-ideal characteristics.

1.2 Design

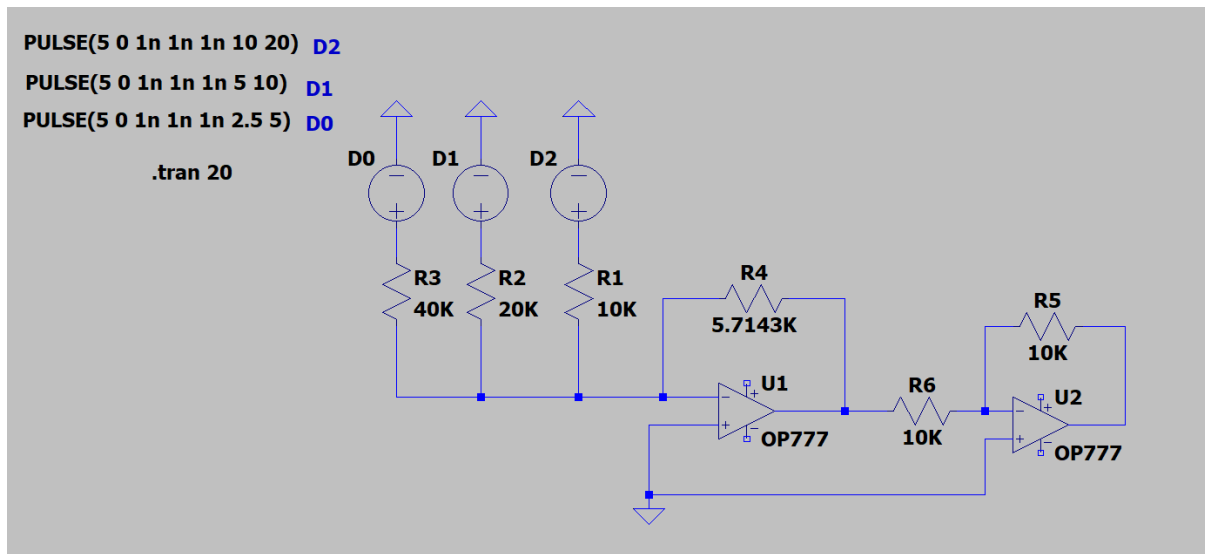


Figure 1: Circuit diagram

As previously mentioned, a simple op-amp based DAC under negative feedback with 1 resistor assigned to each binary digit was used. This specific binary-weighted resistor topology was used since it provides the most simplicity and clarity in the design for demonstrative purposes. More complex design such as a resistor ladder network will not be required as exact precision is not necessary due to the large voltage range output and minimal number of bits to convert (low resolution).

The binary-weighted resistor DAC was implemented using an op-amp (LMC662CN) configured as an inverting summing amplifier in series with a unity gain inverting amplifier.

Each digital input (D2, D1, D0) can be wither 0V (logic low) or 5V (logic high). These inputs are then connected to the inverting input on the op-amp through resistors, with a feedback resistor connected between the output and inverting terminal. The value of the resistor must be assigned and calculated in a specific way. Each input will be connected to a weighted ladder of resistors that are scaled by powers of 2, with the least significant bit (LSB) having the highest resistance (lowest V_{in}), and the most significant bit (MSB) having the lowest resistance (highest V_{in}). The output voltage signal for each corresponding bit code would then be inverted using another amplifier to keep it within the range of 0 to 5V.

This circuit was found from external research and mainly from this source [1].

1.3 Calculations

Using KCL at the negative terminal of the summing op-amp, the transfer function equation can be calculated:

$$V_{-,1} = \frac{-R_f}{R} \left[V_2 + \frac{V_1}{2} + \frac{V_0}{4} \right]$$

Performing KCL again at the negative terminal of the inverting op-amp:

$$V_{out} = -V_{-,1}$$
$$\therefore V_{out} = \frac{R_f}{R} \left[V_2 + \frac{V_1}{2} + \frac{V_0}{4} \right]$$

The maximum voltage for each input and output must be equal to 5V:

$$Max: V_2 = V_1 = V_0 = V_{out} = 5V$$

$$\therefore 5 = 5 \frac{R_f}{R} \left[1 + \frac{1}{2} + \frac{1}{4} \right]$$
$$\Rightarrow R_f \left(\frac{7}{4R} \right) = 1$$

$$\therefore R_f = \frac{4R}{7}$$

Hence the value of R_f must follow this constraint for a maximum output of 5V. Since $R = 10K\Omega$ in our circuit:

$$\therefore R_f = \frac{40}{7} K\Omega \approx 5.714K\Omega$$

Note that this can be approximately constructed using a 1.5K, 1.2K, and 3K resistors in series.

Expected ideal values:

$$\begin{aligned} 000 &\rightarrow 0V \\ 001 &\rightarrow 0.714V \\ 010 &\rightarrow 1.429V \\ 011 &\rightarrow 2.143V \\ 100 &\rightarrow 2.857V \\ 101 &\rightarrow 3.571V \\ 110 &\rightarrow 4.286V \\ 111 &\rightarrow 5.000V \end{aligned}$$

Hence the DAC has an approximate step size of 0.714V

2. Testing on AD2

The designed circuit was built on a breadboard using op-amp supply voltages of +5 and -5 volts.

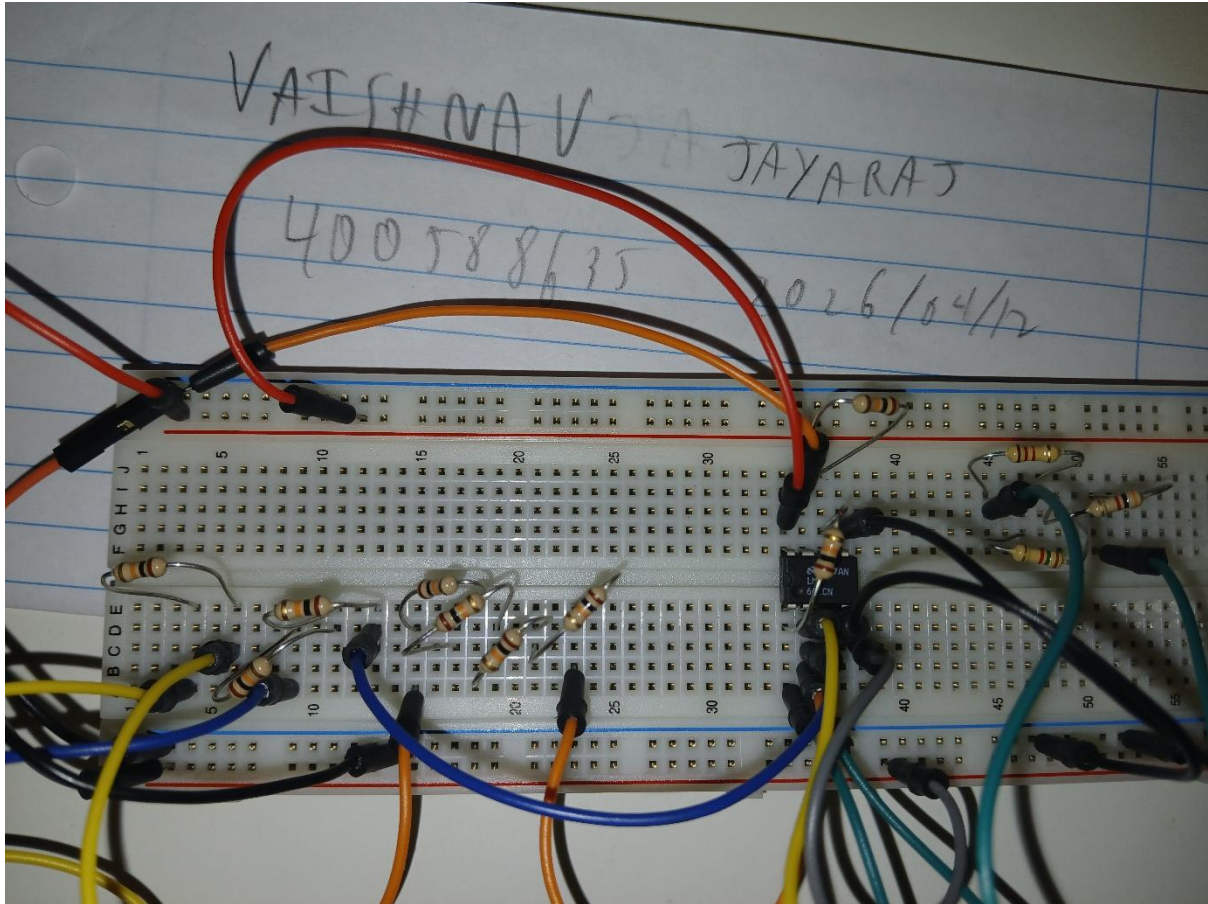


Figure 2: Pic of physical circuit

2.1 Results

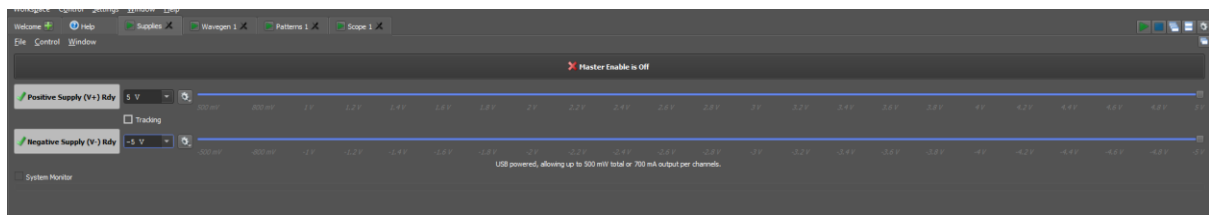


Figure 3: DC op-amp power supply

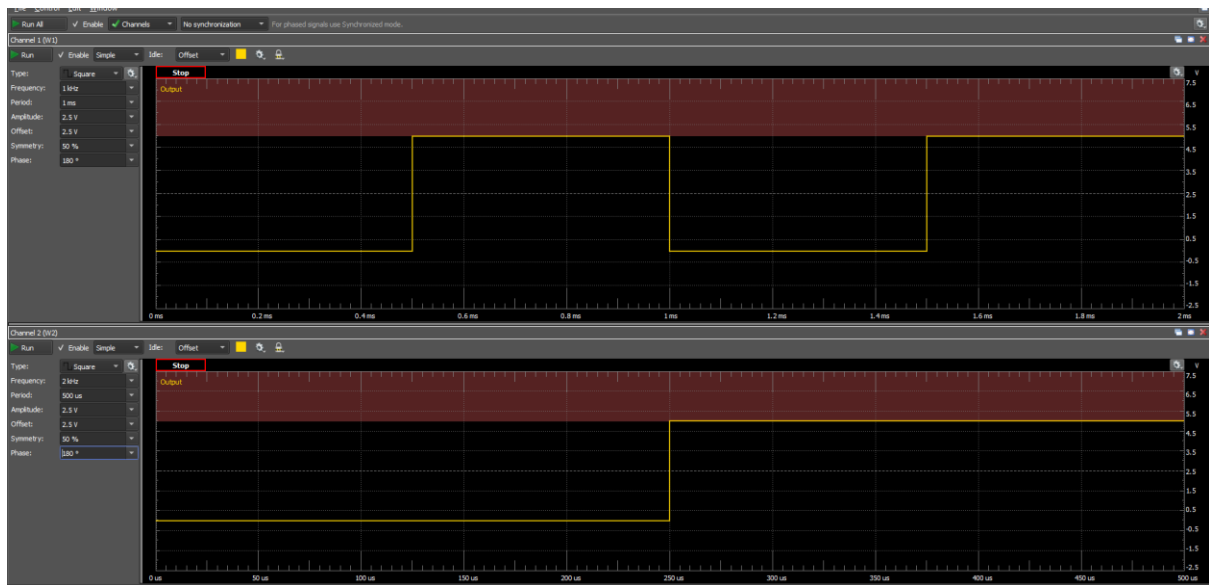


Figure 4: 5V signal inputs

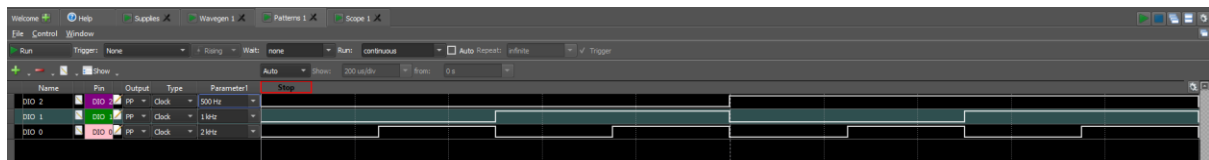


Figure 5: Digital IO inputs (3.3V)



Figure 6: Output with $MSB=0$

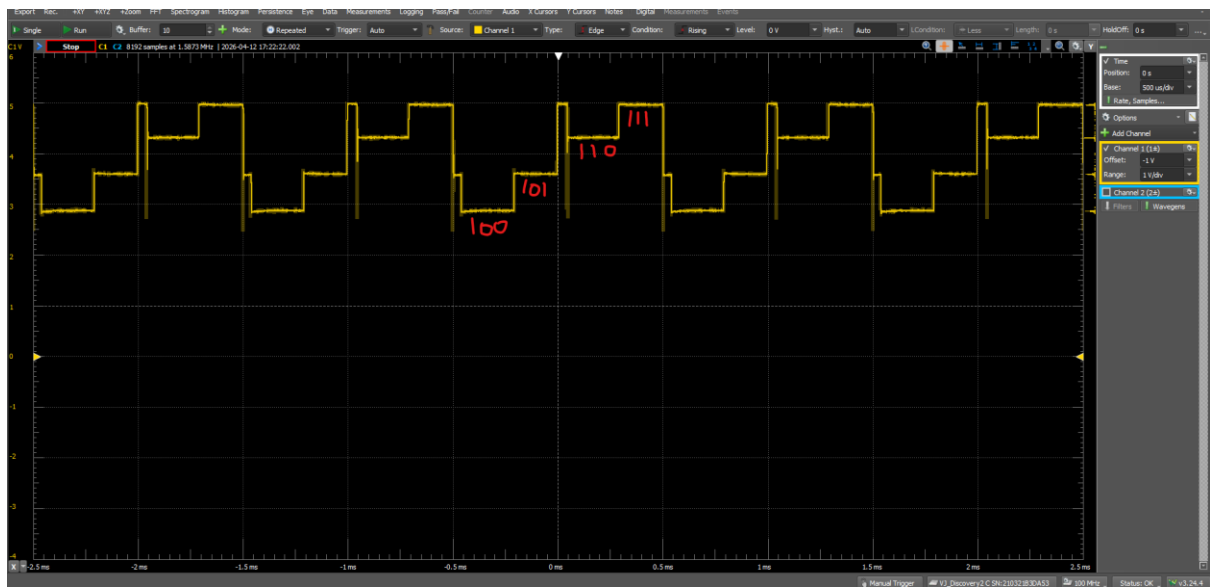


Figure 7: Output with $MSB=1$



Figure 8: Output using digital I/O patterns (3.3V)

The constructed circuit was analysed, and its results were recorded and captured in the images above.

Due to the AD2's voltage and wavegen limitations, all binary codes could not be produced in one graph. Hence, the graphs were split in two one with the MSB=0 (D2 Grounded) and another with the MSB=1 (D2=5V). For both graphs D0 and D1 were square waves with a voltage from 0-5V with D0 having twice the frequency than D1 (to cycle through all binary digits). Both Figures 6 and 7 clearly show each output level associated with their respective binary code. This proves that the circuit created creates a unique analog output for each binary code received.

For more simpler and clearer visualization of the complete signal, a patterns tab was also run with each input connected to a digital IO port that will produce all the binary digits sequentially. However, the logic high voltage for each input in this case is 3.3V and not 5V.

2.2 Gain Error

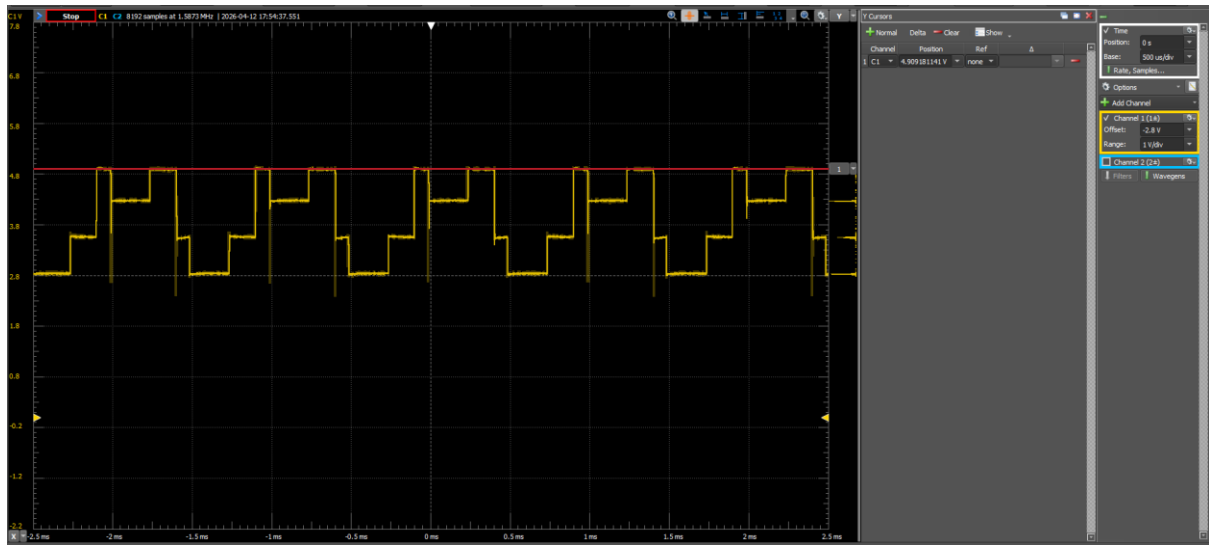


Figure 9: Max recorded voltage

From the graph the maximum gain error can be calculated by comparing the expected maximum voltage to the recorded one.

$$E_{Gain} = \frac{V_{measured} - V_{expected}}{V_{expected}} \times 100\%$$

The maximum recorded voltage was approximately 4.909V.

$$\begin{aligned} \therefore E_{Gain} &= \frac{4.909 - 5.000}{5.000} \times 100\% \\ \therefore E_{Gain} &\approx \mathbf{1.82\%} \end{aligned}$$

2.3 Maximum differential non-linearity

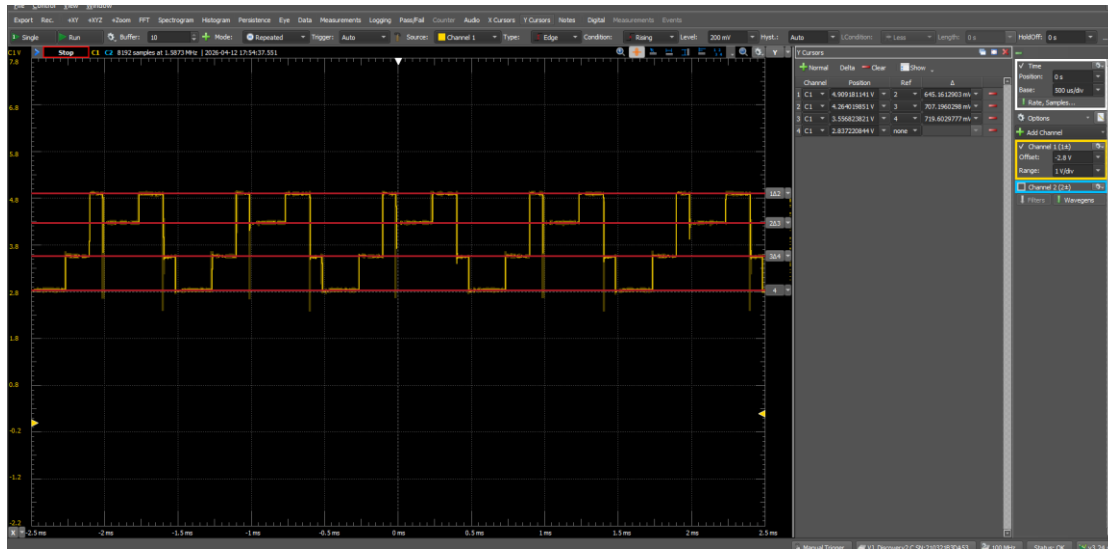


Figure 10: Step values when MSB=1

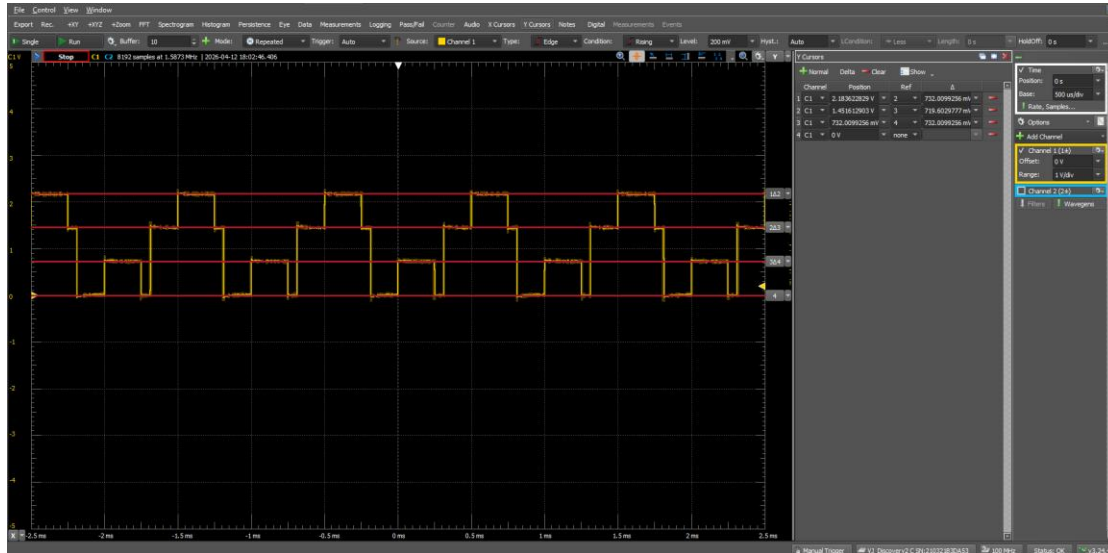


Figure 11: Step values when MSB=0

In an ideal 3-bit DAC with a 5V full scale, each LSB bit represents 0.714V. The Differential non-linearity (DNL) quantifies the deviation in step size for each measured step. The maximum of these deviations (DNL_{max}) will define the worst-case non-linearity. A high DNL_{max} indicated to more unevenly spaced levels. In our case, the maximum deviation in step size was from the 110 to the 111 bit, with a step size of 0.645V. Hence the DNL_{max} can be calculated:

$$DNL_{max} = \frac{V_{dev,max} - V_{step}}{V_{step}} \times 100\%$$

$$\therefore DNL_{max} = \frac{0.645 - 0.714}{0.714} \times 100\% \approx 9.66\%$$

2.4 Offset Error

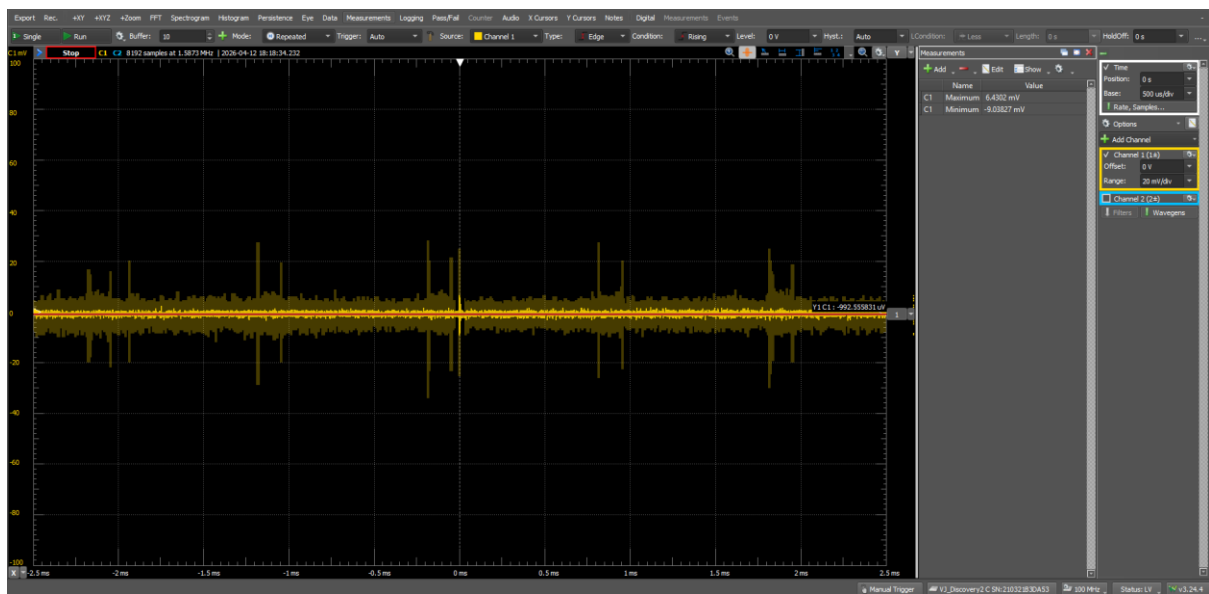


Figure 12: Voltage output when bits are 000

The offset error is defined as the output deviation from 0V when the DAC's binary code is zero (000). The offset can hence be measured by driving all the inputs to logical low (0V) and recording the output voltage. In our design, when all inputs were 0, there was a maximum output voltage of 6.4302 mV and a minimum output voltage of -9.03827 mV. This measured voltage is mainly due to external noises in the circuit, leakage currents from connections, and the use of non-ideal components.

3. Discussion

From the resulting graphs and outputs recorded, it can be clearly seen that a DAC design was successfully created with an inverting summing amplifier, and an inverting unity gain amplifier, that can assign any 3 bit digital binary code to a specific analog dc value. The designed circuit had attenuation loss of approximately 1.82%, and a maximum step deviation of approximately 9.66%. Most of the errors in this design can be attributed to the AD2's limitation and op-amp non idealities. The AD2 can only output a maximum of 5V, with majority of its outputs not being true 5V. Furthermore we are only limited to 2 wavegens for the analysis. The net amount of voltage required for the circuits operation overall exceeded the AD2's capabilities, resulting in lower quality wavegen and power outputs. This can be clearly seen in the graphs were there is faulty output level switching between binary codes. Errors can also be attributed to the use of non-ideal op-amps, as well as external conditions.

This specific design was selected mainly due to its simplicity and direct mapping of digital inputs to analog outputs. This circuit provides a more basic and robust way of converting bits as well as using minimal components and being easier to troubleshoot.

Alternative DAC designs mainly use R-2R ladder designs, which offer much better resolution and precision in the final output. They use many resistors matched to each other so that they can achieve step sizes to mV accuracies. These kinds of architectures are more commonly found in large scale commercial systems where the conversion of 5 or more-bit binary codes are required. It is extremely hard to implement this design given the current equipment and component limitations given to us. However, this design is much more complex, requiring several components and troubleshooting steps if required.

References:

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- [3] "Digital-to-Analog Converter - an overview | ScienceDirect Topics," *www.sciencedirect.com*. <https://www.sciencedirect.com/topics/engineering/digital-to-analog-converter>
- [4] ElectronicsTutorials, "R-2R Digital-to-Analogue Converter (R-2R DAC)," *Basic Electronics Tutorials*, Oct. 13, 2020. <https://www.electronics-tutorials.ws/combination/r-2r-dac.html>